

Problem 1

3 shirts x 4 pairs of pants = 12 pairs of shirt & pant

Problem 2

$$\begin{array}{c} \text{1 ch} \\ \hline 0-9 \\ \hline 10 \end{array} \times \begin{array}{c} \text{2 ch} \\ \hline a-z(x0) \\ \hline 25 \end{array} \times \begin{array}{c} \text{3 ch} \\ \hline 0-9 \\ \hline 10 \\ \hline a-z \\ \hline (9+26-1) \end{array} \times \begin{array}{c} \text{4 ch} \\ \hline 0-9 \\ \hline 10 \\ \hline a-z \\ \hline (9+26-1) \end{array} \times \begin{array}{c} \text{5 ch} \\ \hline 0-9 \\ \hline 10 \\ \hline a-z \\ \hline (9+26-1) \end{array} \times \begin{array}{c} \text{6 ch} \\ \hline 0-9 \\ \hline 10 \\ \hline a-z \\ \hline (9+26-1) \end{array} \times \begin{array}{c} \text{7 ch} \\ \hline 0-9 \\ \hline 10 \\ \hline a-z \\ \hline (9+26-1) \end{array} =$$
$$= 10 \times 25 \times (34^5) = 38250 = \underline{\underline{13130468750}}$$

Problem 3

$$S = 1 + 2 + \dots + n-1 + n$$

$$S = n + (n-1) + \dots + 2 + 1$$

$$2S = (n+1) + (n+1) + \dots + (n+1) + (n+1)$$

$$S = \frac{n(n+1)}{2}$$

$\Rightarrow X$

$$a) 25 \times (25-1) \times (25-2) \times (25-3) = \underline{303600} \leftarrow \frac{25!}{(25-4)!}$$

$$b) 25 \times 25 \times 25 \times 25 = 25^4 = \underline{390625}$$

Problem 4

5 letters

$$0 < \text{word.length} \leq 3 \Rightarrow$$

1 ch words

$$1 \times 5$$

2 ch words

$$(1 \times 5) \times (1 \times 5)$$

$$= 5^2 = 25$$

3 ch words

$$(1 \times 5) \times (1 \times 5) \times (1 \times 5)$$

$$= 5^3 = 125$$

$$= 5 + 25 + 125 = \underline{\underline{155 \text{ words}}}$$

Problem 5

$$\begin{array}{ccccccc}
 \times & \text{BBBB} & \text{GGG} & \Rightarrow & \text{BBBGGG} & \text{BBBGGG} & \text{BBBGGG} \\
 & \underbrace{\quad} & & & & & \\
 & \text{B} & & & \text{BBGGGG} & \text{BBGGGG} & \text{BBGGGG} \\
 & & & & & & \\
 & & & & \text{BBGGGG} & \text{BBGGGG} & \text{BBGGGG}
 \end{array}
 \left. \vphantom{\begin{array}{c} \text{BBBB} \\ \text{GGG} \end{array}} \right\} \begin{array}{l} \text{total} \\ 6 \\ \text{conf.} \end{array}$$

Complementary

$$\begin{array}{c}
 \text{BGGGBGB} \leftarrow \text{no BB} \\
 \underbrace{\quad\quad\quad} \quad \underbrace{\quad\quad} \\
 4! = 24 \quad 3! = 6 \rightarrow 24 \times 6 = 144
 \end{array}$$

$$7! = 5040 - 144 = \underline{4896}$$

Problem 6

$$x_i = \{1, 2, 3, 4, 5\}$$

$$x_1 = 5, x_2 = 4, \dots$$

$$\downarrow \text{choices} \quad (x_i \neq x_{i-1})$$

$$\begin{array}{ccccccc}
 x_1 & x_2 & x_3 & x_4 & x_5 & x_6 & x_7 \\
 \Rightarrow & 5 & \times & 4 & \times & 4 & \times & 4 & \times & 4 & \times & 4 & \times & 4 & \times & 4 & \times & 4
 \end{array}
 = 5 \times 4^6 = 5 \times 4096 = \underline{20480}$$

Problem 7

$$1. \frac{20!}{(20-3)!} = \frac{20!}{17!} = 6840$$

$$2. 2 \times 3 = 6 \Rightarrow 6 \times (20-2) = 108$$

$$3. 6840 - 108 = \underline{6732}$$

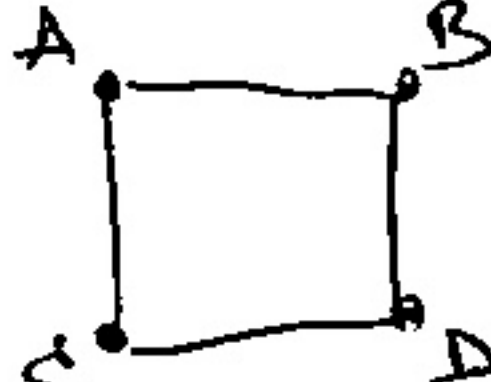
Problem 8

$$\begin{array}{c}
 \text{PAPA} \\
 \underbrace{\quad\quad} \quad \underbrace{\quad\quad} \\
 2! \quad 2!
 \end{array}
 \rightarrow 4! = 24 \Rightarrow \frac{4!}{2!2!} = \frac{24}{4} = \underline{6}$$

Problem 9

① $\frac{n!}{(n-2)! \cdot 2!} = \text{total} - n \text{ sides} = k$

Annotations:
 $n!$ ← all permutations
 $(n-2)! \cdot 2!$ ← order doesn't matter
 ↑ need only 2 pairs
 k ← number of diagonals

② Ex:  $A: n-1 = 4-1 = 3$
 $B: n-2 = 4-2 = 2$
 $C: n-3 = 4-3 = 1$
 $6 - 4 \text{ sides} = 2 \text{ diagonals}$

Ans: $\frac{n(n-1)}{2} - n = \frac{n(n-1)}{2} - \frac{2n}{2} = \frac{n(n-1)-2n}{2} = \frac{n^2-n-2n}{2} = \frac{n^2-3n}{2} = \frac{n(n-3)}{2}$

Problem 10

$\frac{48!}{(48-6)! \cdot 6!} = \frac{48 \times 47 \times 46 \times 45 \times 44 \times 43}{6!} = 12271512$

Annotations:
 $6!$ ← order doesn't matter

Problem 11

1. $\frac{12!}{(12-5)! \cdot 5!} = \frac{12 \times 11 \times 10 \times 9 \times 8}{5!} = 792$

2. $\frac{10!}{(10-3)! \cdot 3!} = 120$

Annotations:
 bob, yogi, and 3 others (12-2=10 players)

3. $792 - 120 = 672$

Problem 12

$$\binom{n}{k} \cdot x^{n-k} \cdot y^k \Rightarrow \binom{6}{k} \cdot x^{6-k} \cdot (2y^2)^k$$

$$k=4 \Rightarrow \binom{6}{4} \cdot x^{6-4} \cdot (2^4 \cdot y^8)$$

$$C(6,4) = \frac{6!}{4!(6-4)!} = 15$$

$$= \frac{15}{1} \cdot x^2 \cdot \frac{16}{1} \cdot y^8$$

$$= \frac{15}{1} \times \frac{16}{1}$$

$$\frac{90}{1}$$

$$+ 15$$

$$\underline{\underline{240}} x^2 y^8$$